

Task 1 :

1. AWS RDS is a storage system that we use to store and manipulate Structured Data using a query language

It supports various databases like MySQL, Microsoft SQL, Maria DB etc..

2. Data Lake is a storage class where we are going to store the data in its raw format

In a data lake we can store structured, Unstructured and Semi-structured Data

3. Amazon Glacier is one of the storage classes in Amazon S3, where we store the rarely used data or the data that we never use, but we need to store the data.

In such scenarios we use Amazon Glacier so that the cost for storage will be optimized and the data can be retrieved when we need to use it.

Task 2:

Let us consider Netflix as an example

1. AWS RDS

Used to store structured data and connect to a database using queries

Example Use Case:

We use the RDS to store the credentials of each user in the form of tables.

2. Data Lake

AWS Data Lake is used to store the raw Data structured, unstructured and semi structured Data.

Example Use Case:

In a Data Lake Netflix stores the data of templates and ratings data.

3. Amazon Glacier :

This stores the rarely used data or the archival data.

Use Case :

Watch History Data before few years

Task 3:

I have used the existing RDS Instance and existing DataBase.

DataBase Name: ecommerce

I have taken CreateTable.sh file to perform CRUD operations and Execute the script file and

Checked the database in SQL Server Management Studio.

created a table named student

performed INSERT, DELETE and UPDATE operations on the table

Task 4:

1. If companies want to install the Data Bases Manually they need to Have multiple servers and It requires a lot of cost and more complex to install many system and servers so, if they choose AWS RDS they can use the servers based on the requirement and the cost will be optimized as AWS offers Pay-as - you go service so that the cost will be optimized and resources can be utilized based on the requirement.

2.DataLake stores the data in the raw format. we don't need to preprocess the data before storing in a dataLake. we can preprocess the Data while we want to use it .
We can store Structured, Unstructured and Semi Structured Data into a DataLake.

3. Amazon Glacier is cheaper than other storage services because it stores the Data in the compressed format and we only store the archival data into Amazon Glacier. as it stores the rarely used data it stores in compression format. So, when an user requires it they can use it after extracting it .